

2. Constructing pose maps

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1. Use slam_toolbox to build raster maps and pose maps

[!NOTE]

- The road network .geojson file must be preceded by at least one raster map.
- Pose maps are designed to enable rapid relocalization during waypoint marking and navigation, improving the accuracy of global positioning and waypoint marking.

2. Building a raster map

- Start the terminal of Humble version of Docker

```
start_muto_humble
```

- Terminal input command

```
ros2 launch yahboomcar_nav map_slam_toolbox_launch.py
```

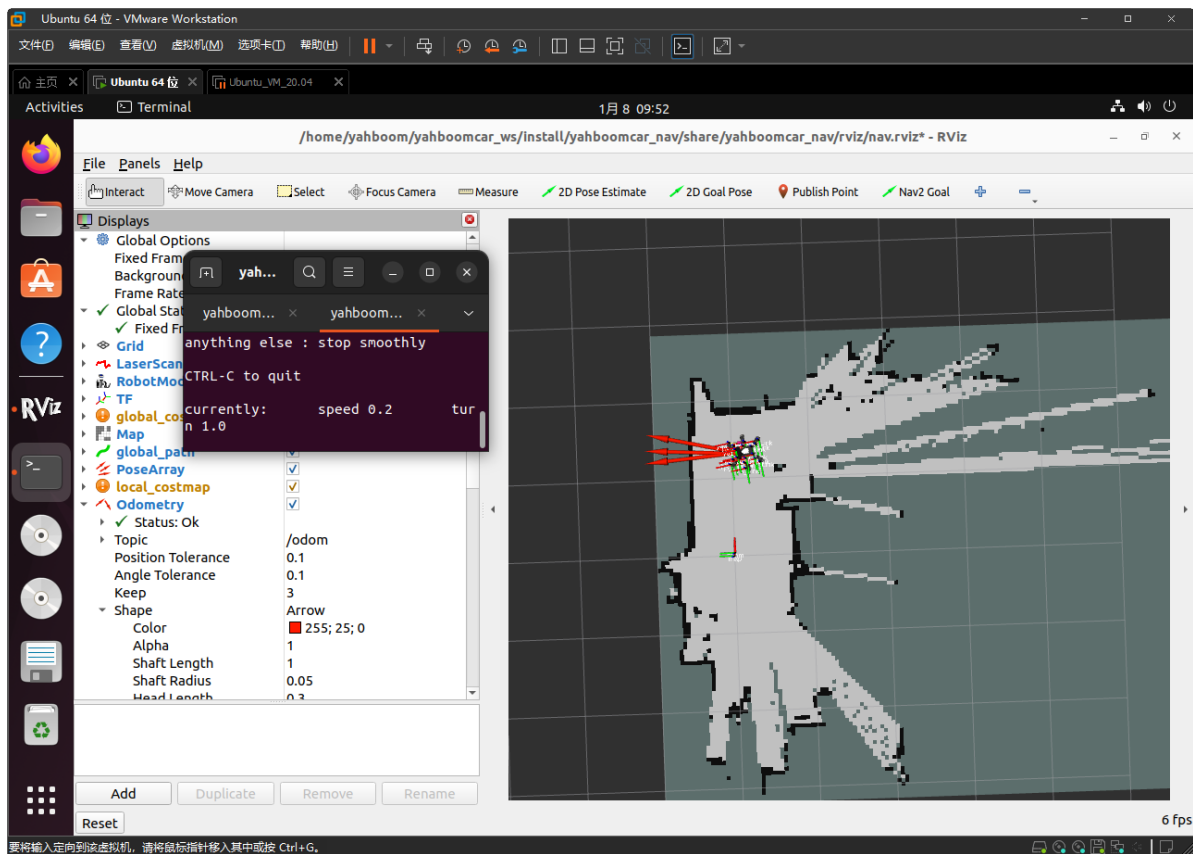
Open RVIZ in a virtual machine or terminal.

```
ros2 launch yahboomcar_nav display_nav_launch.py
```

- Restart the keyboard control node (or use a gamepad).

```
ros2 run yahboomcar_ctrl yahboom_keyboard
```

- Controlling vehicle movement for mapping



- Save raster map

```
ros2 service call /slam_toolbox/save_map slam_toolbox/srv/SaveMap "{name: {data:
'/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/road_map'}}"
```

[!TIP]

- `name` is used to specify the **map name** of the saved raster map; in this case, it's `road_map`.

After execution, we can check if the save was successful (we can see that a `road_map.pgm` and a `road_map.yaml` file have been added).

```
root@raspberrypi:/# ros2 service call /slam_toolbox/save_map
slam_toolbox/srv/SaveMap "{name: {data:
'/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/road_map'}}"
requester: making request:
slam_toolbox.srv.SaveMap_Request(name=std_msgs.msg.String(data='/root/yahboomcar
_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/road_map'))

response:
slam_toolbox.srv.SaveMap_Response(result=0)

root@raspberrypi:/# ll
/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/
total 252
drwxr-xr-x 2 1000 1000 4096 Jan 8 01:53 ./
drwxr-xr-x 9 1000 1000 4096 Jan 4 13:14 ../
-rw-r--r-- 1 root root 22763 Jan 8 01:53 road_map.pgm
```

```
-rw-r--r-- 1 root root 125 Jan 8 01:53 road_map.yaml
-rw-r--r-- 1 1000 1000 159459 Jan 4 13:14 test.pgm
-rw-r--r-- 1 1000 1000 181 Jan 4 13:14 test.yaml
-rw-r--r-- 1 1000 1000 51885 Jan 4 13:14 yahboomcar.pgm
-rw-r--r-- 1 1000 1000 126 Jan 4 13:14 yahboomcar.yaml
root@raspberrypi:/#
```

- Save the slam_toolbox pose map

```
ros2 service call /slam_toolbox/serialize_map
slam_toolbox/srv/SerializePoseGraph "{filename:
'/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/road_map'}"
```

[!TIP]

- `map_type` stores the map type. The default is `gridmap`, which stores a raster map. Selecting `posegraph` stores a pose map, which is a `slam_toolbox`.

The terminal displays `slam_toolbox.srv.SerializePoseGraph_Response(result=0)`, indicating that the pose map has been successfully saved.

The terminal content is as follows:

```
root@raspberrypi:/# ros2 service call /slam_toolbox/serialize_map
slam_toolbox/srv/SerializePoseGraph "{filename:
'/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/road_map'}"
requester: making request:
slam_toolbox.srv.SerializePoseGraph_Request(filename='/root/yahboomcar_ros2_ws/y
ahboomcar_ws/src/yahboomcar_nav/maps/road_map')

response:
slam_toolbox.srv.SerializePoseGraph_Response(result=0)

root@raspberrypi:/# ll
/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/
total 18260

drwxr-xr-x 2 1000 1000 4096 Jan 8 01:54 ./
drwxr-xr-x 9 1000 1000 4096 Jan 4 13:14 ../
-rw-r--r-- 1 root root 157945 Jan 8 01:54 road_map.data
-rw-r--r-- 1 root root 22763 Jan 8 01:53 road_map.pgm
-rw-r--r-- 1 root root 18278895 Jan 8 01:54 road_map.posegraph
-rw-r--r-- 1 root root 125 Jan 8 01:53 road_map.yaml
-rw-r--r-- 1 1000 1000 159459 Jan 4 13:14 test.pgm
-rw-r--r-- 1 1000 1000 181 Jan 4 13:14 test.yaml
-rw-r--r-- 1 1000 1000 51885 Jan 4 13:14 yahboomcar.pgm
-rw-r--r-- 1 1000 1000 126 Jan 4 13:14 yahboomcar.yaml
root@raspberrypi:/#
```

